Stage 1 | Sequence & Positional Encoding

41-bp DNA Sequence

41-bp Window

Target Site C / A*

Token Embedding

DNA Vocabulary (d = 128)

Embed | Scale | Dropout

Token Matrix

A1
C2
G20
C/A21*
T41
[PAD] 0

41 × 128

Positional Encoding

Sinusoidal PE

Position Phase

Position-Aware Input

Token + Positional Fusion

Sin | Cos | Norm

Encoded Sequence

t=1
t=2
t=20
t=21*
t=41
[PAD] 0

41 × 128

Stage 2 | Time-Frequency Front-End

Temporal-Spatial Branch

Dilated TCN + Multi-Scale CNN

TCN | MS-CNN | GeLU

Multi-Scale Motif Features

Temporal

41 × 128

Adaptive Gated Fusion

Frequency-Wavelet Branch

Learnable VMD + Orthogonal Wavelet

VMD | Wavelet | Shrink

Wavelet-Spectral Features

Spectral

41 × 128

Stage 3 | Dual-Path Context Backbone & SE Fusion

Sequential BiLSTM Pathway

Forward

Backward

2-Layer BiLSTM

BiLSTM Out

41 × 256

Global Self-Attention Pathway

Query

Key / Val

8-Head MHSA

Attention Out

41 × 128

Sequential Context

41 × 256

SE Channel Fusion

Channel Attention

Global Context

41 × 128

Stage 4 | Target-Anchored Pooling & Ensemble Readout

Context

41 × 384

1. Deep MLP Classification Branch

Methylation Score

Classifier Head

2. Positional Attention & Latent Readout

Attention

Latent Space

Target Readout

Context Vector

5-Fold Ensemble & Training

Cross-Entropy Loss

5-Fold Soft-Voting

Ensemble Probability

AdamW + Cosine LR

Stage 5 | Epigenomic Attribution & Generalization

Motif Recovery

Center Peak

A*TGC

GAGG

-20 -5 0* +5 +20

Core Cis-Element Loci

(±5 bp Target Window)

Domain Attribution

Temporal

Local Motif

Core Motif Loci

Spectral

Wavelet Flank

Flanking Envelope

Time-Frequency

Domain Separation

Species Transfer

4mC (4 Species)

6mA (11 Species)

Cross-Species AUC

Noise Robustness

15-Species Benchmark

Generalization